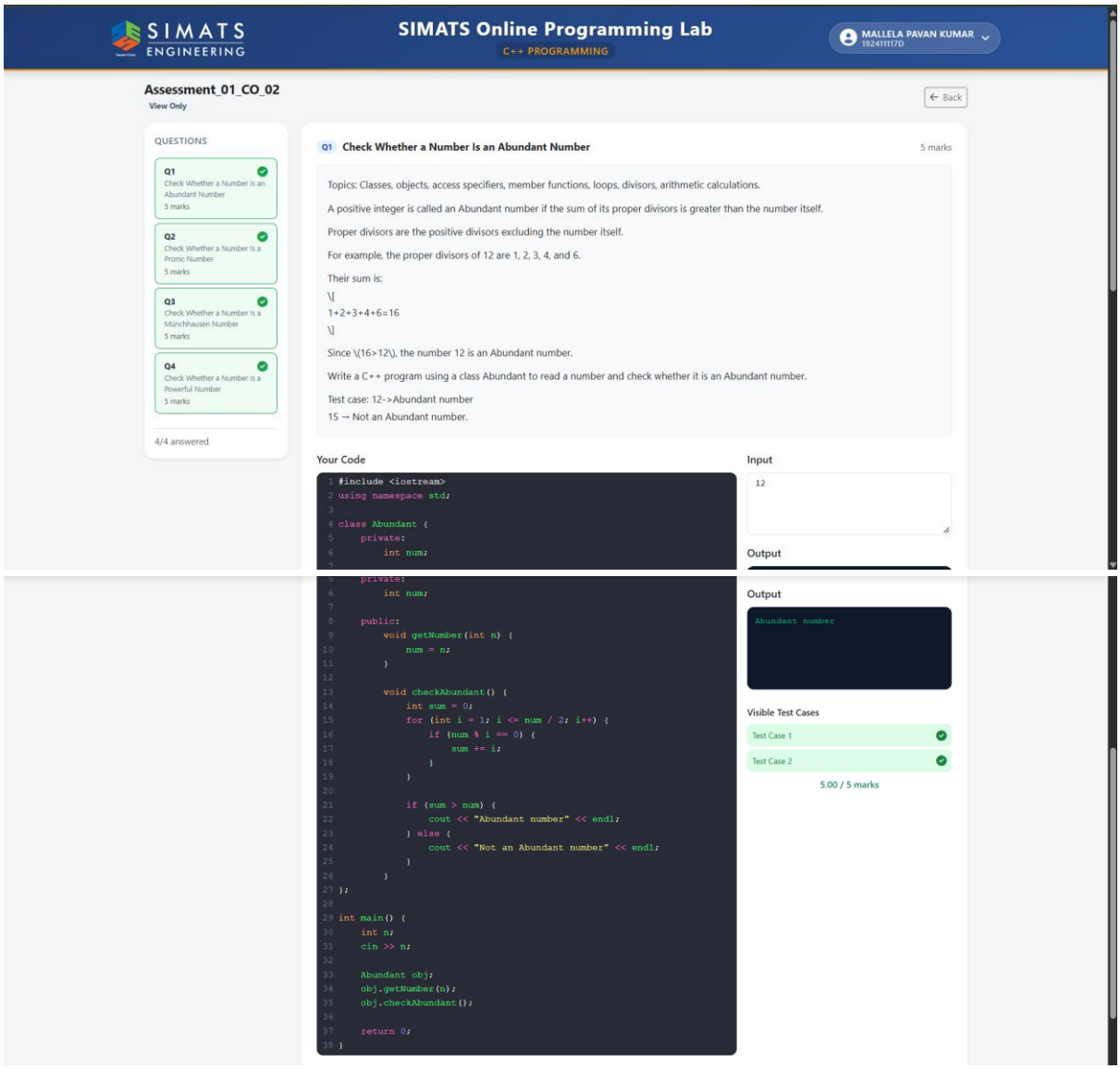
M.Pavan kumar 192411117

DSA0120 SLOT-D




```cpp
#include <iostream>
using namespace std;

class Abundant {
    private:
        int num;

    public:
        void getNumber(int n) {
            num = n;
        }

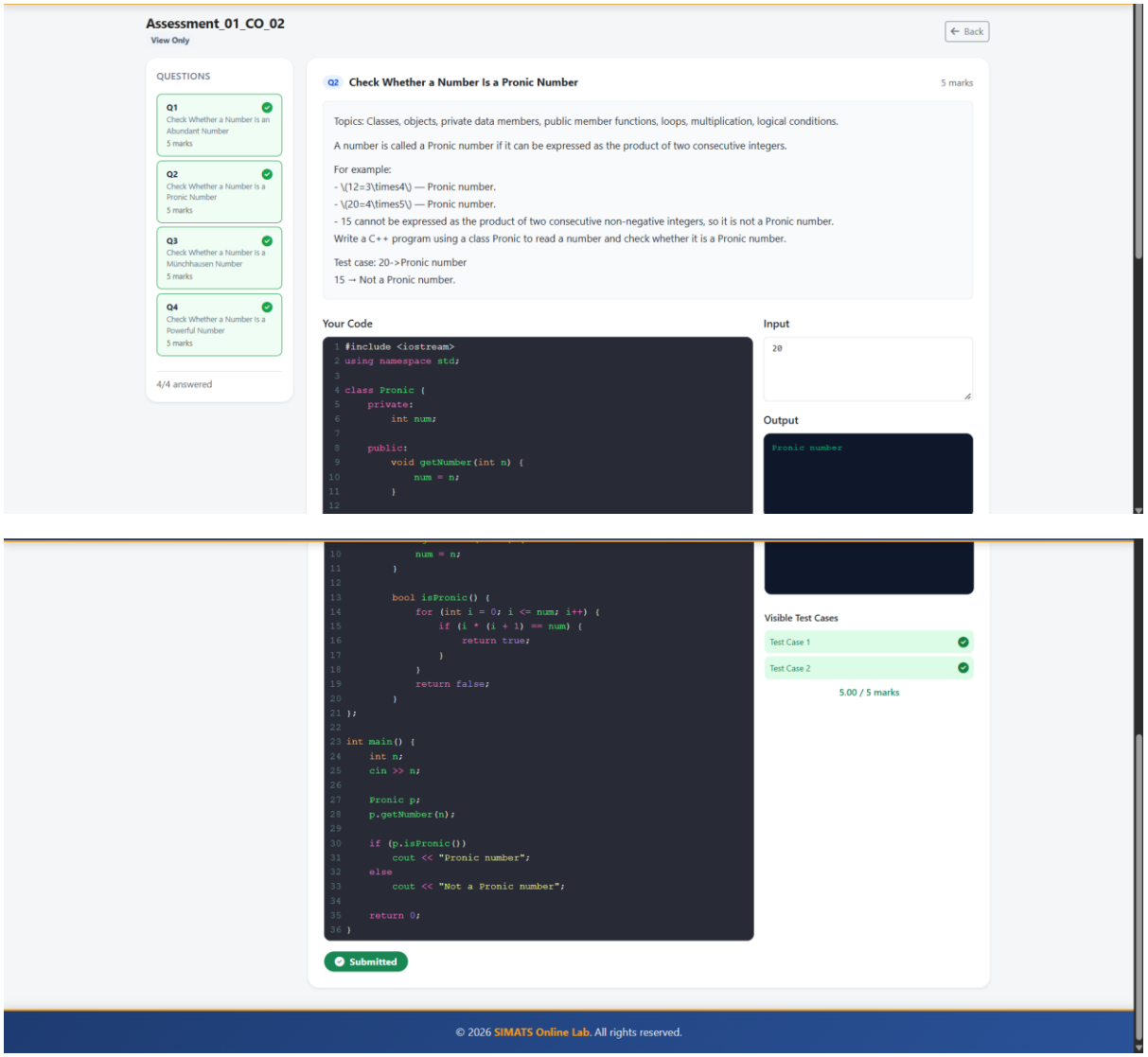
        void checkAbundant() {
            int sum = 0;
            for (int i = 1; i <= num / 2; i++) {
                if (num % i == 0) {
                    sum += i;
                }
            }

            if (sum > num) {
                cout << "Abundant number" << endl;
            } else {
                cout << "Not an Abundant number" << endl;
            }
        }
};

int main() {
    int n;
    cin >> n;

    Abundant obj;
    obj.getNumber(n);
    obj.checkAbundant();

    return 0;
}
```



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QUESTIONS

Q1 ✓
Check Whether a Number Is an Abundant Number
5 marks

Q2 ✓
Check Whether a Number Is a Pronic Number
5 marks

Q3 ✓
Check Whether a Number Is a Münchhausen Number
5 marks

Q4 ✓
Check Whether a Number Is a Powerful Number
5 marks

4/4 answered

Q3 Check Whether a Number Is a Münchhausen Number

5 marks

Topics: Classes, objects, access specifiers, member functions, digit extraction, loops, and exponentiation.

A Münchhausen number is a number that is equal to the sum of its digits, with each digit raised to the power of itself. For this program, consider \(0^0=0\).

For example, the number 1 is a Münchhausen number because \(1^1=1\).

Write a C++ program using a class Munchhausen to read a number and check whether it is a Münchhausen number.

Test case: 1-> Munchhausen number
10 → Not a Munchhausen number.

For 10, the sum is \(1^1+0^0=1\), which is not equal to 10.

Your Code

```cpp
#include<iostream>
#include <cmath>
using namespace std;

class Munchhausen {
    private:
        int num;

    public:
        void getNumber(int n) {
            num = n;
        }

        int power(int base, int exp) {
            if (base == 0) return 0;
            int result = 1;
            for (int i = 0; i < exp; i++) {
                result *= base;
            }
            return result;
        }

        bool isMunchhausen() {
            int temp = num;
            int sum = 0;

            if (temp == 0) {
                return true;
            }

            while (temp > 0) {
                int digit = temp % 10;
                sum += power(digit, digit);
                temp /= 10;
            }

            return (sum == num);
        }
};

int main() {
    int n;
    cin >> n;

    Munchhausen m;
    m.getNumber(n);

    if (m.isMunchhausen()) {
        cout << "Munchhausen number" << endl;
    } else {
        cout << "Not a Munchhausen number" << endl;
    }
```

Input

```
1
```

Output

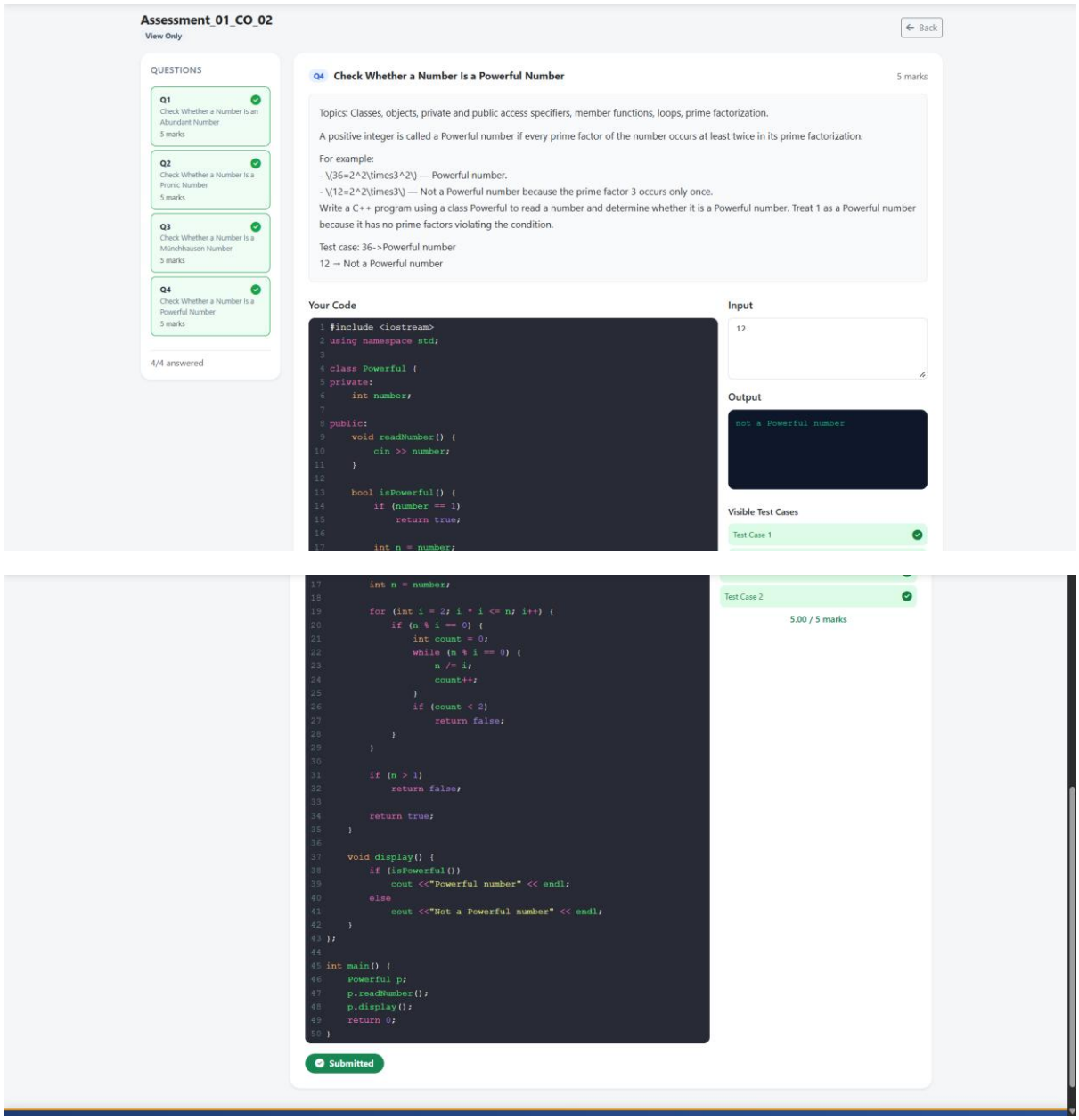
```
Munchhausen number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

5.00 / 5 marks


Assessment_01_CO_02
View Only
Back
QUESTIONS
Q1
Check Whether a Number is an Abundant Number
5 marks
Q2
Check Whether a Number is a Pronic Number
5 marks
Q3
Check Whether a Number is a Münchhausen Number
5 marks
Q4
Check Whether a Number is a Powerful Number
5 marks
4/4 answered
Q4 Check Whether a Number Is a Powerful Number
5 marks
Topics: Classes, objects, private and public access specifiers, member functions, loops, prime factorization.
A positive integer is called a Powerful number if every prime factor of the number occurs at least twice in its prime factorization.
For example:
- \(36=2^2\times3^2\) — Powerful number.
- \(12=2^2\times3\) — Not a Powerful number because the prime factor 3 occurs only once.
Write a C++ program using a class Powerful to read a number and determine whether it is a Powerful number. Treat 1 as a Powerful number because it has no prime factors violating the condition.
Test case: 36->Powerful number
12 → Not a Powerful number
Your Code
#include <iostream>
using namespace std;
class Powerful {
private:
int number;
public:
void readNumber() {
cin >> number;
}
bool isPowerful() {
if (number == 1)
return true;
int n = number;
for (int i = 2; i * i <= n; i++) {
if (n % i == 0) {
int count = 0;
while (n % i == 0) {
n /= i;
count++;
}
if (count < 2)
return false;
}
}
if (n > 1)
return false;
return true;
}
void display() {
if (isPowerful())
cout <<"Powerful number" << endl;
else
cout <<"Not a Powerful number" << endl;
}
};
int main() {
Powerful p;
p.readNumber();
p.display();
return 0;
}
Input
12
Output
not a Powerful number
Visible Test Cases
Test Case 1
Test Case 2
5.00 / 5 marks
Submitted